Sedrick Keh

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Research Experience

- **Anthropic** Apr. 2026 - Present
  Member of Technical Staff
  - **Pre-training:** Large-scale data creation for multimodal models.

- **Toyota Research Institute** May 2023 - Mar. 2026
  Machine Learning Research Scientist
  - **Pre-training:** Data mixtures for pre-training; attention mechanism modifications; model evaluation. Helped pre-train 1B to 7B models on trillions of text tokens [7] [9] [12].
  - **Post-training:** Reasoning; supervised fine-tuning (SFT); synthetic data generation [4] [8]. Co-first author of OpenThoughts [3].
  - **RLHF:** Investigated reinforcement learning with AI feedback (RLAIF) [10].
  - **Multimodal Models:** Pre-training for vision-language models (VLMs) [6]; LLM planning for robotics applications; training and evaluating vision-language-action (VLA) models [2] [5] [22].

- **Carnegie Mellon University – Language Technologies Institute** Sept. 2021 - May 2023
  Advised by Prof. Daniel Fried and Prof. Eduard Hovy
  - **Multimodal Models, Grounded Interaction:** Visually-grounded question generation in a collaborative uncertainty resolution task [13].
  - **Creative and Controllable Text Generation**: Commonsense reasoning and data augmentation strategies to generate figurative language [16] [19] [20] [21].

- **Carnegie Mellon University – Institute for Software Research** Sept. 2021 - May 2023
  Advised by Prof. Fei Fang
  - **AI/NLP for Social Good:** Deployed a multilingual system to automatically extract and classify news articles related to wildlife conservation. Collaboration with international WWF offices [14] [18].

Peer-Reviewed Publications

[1] DataComp-VLM: Improved Open Datasets for Vision-Language Models
*Matteo Farina, Vishaal Udandarao, Thao Nguyen, [...], **Sedrick Keh**, [...], Ludwig Schmidt, Nikhil Parthasarathy (36 authors).*
The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026), Evaluations and Datasets Track. December 2026.

[2] SUAVE: Unified Video-Action Models via Masked Diffusion
*Rhythm Syed, Jean Mercat, **Sedrick Keh**, Kushal Arora, Paarth Shah, Aykut Onol, Mengchao Zhang, Tony Dear.*
Pretrain to Adapt Workshop and G2A Workshop (CoRL 2026). November 2026.

[3] OpenThoughts: Data Recipes for Reasoning Models
Etash Guha, Ryan Marten*, **Sedrick Keh***, Negin Raoof*, Georgios Smyrnis*, Hritik Bansal, Marianna Nezhurina, Jean Mercat, Trung Vu, Zayne Sprague, Ashima Suvarna, Benjamin Feuer, Liangyu Chen, Zaid Khan, Eric Frankel, Sachin Grover, Caroline Choi, Niklas Muennighoff, Shiye Su, Wanjia Zhao, John Yang, Shreyas Pimpalgaonkar, Kartik Sharma, Charlie Cheng-Jie Ji, Yichuan Deng, Sarah Pratt, Vivek Ramanujan, Jon Saad-Falcon, Stutee Acharya, Jeffrey Li, Achal Dave, Alon Albalak, Kushal Arora, Blake Wulfe, Chinmay Hegde, Greg Durrett, Sewoong Oh, Mohit Bansal, Saadia Gabriel, Aditya Grover, Kai-Wei Chang, Vaishaal Shankar, Aaron Gokaslan, Mike A. Merrill, Tatsunori*

Hashimoto, Yejin Choi, Jenia Jitsev, Reinhard Heckel, Maheswaran Sathiamoorthy, Alexandros G. Dimakis, Ludwig Schmidt.
The Fourteenth International Conference on Learning Representations (ICLR 2026), Oral. April 2026.

[4] SkillFactory: Self-Distillation for Learning Cognitive Behaviors
Zayne Sprague, Jack Lu, Manya Wadhwa, ***Sedrick Keh****, Mengye Ren, Greg Durrett.*
The Fourteenth International Conference on Learning Representations (ICLR 2026). April 2026.

[5] A Careful Examination of Large Behavior Models for Multitask Dexterous Manipulation
TRI LBM Team: Jose Barreiros, Andrew Beaulieu, Aditya Bhat, [...], ***Sedrick Keh****, [...], Russ Tedrake (81 authors).*
Science Robotics, Volume 11, Issue 113. April 2026.

[6] Should VLMs be Pre-trained with Image Data?
Sedrick Keh*, Jean Mercat*, Samir Yitzhak Gadre, Kushal Arora, Igor Vasiljevic, Benjamin Burchfiel, Shuran Song, Russ Tedrake, Thomas Kollar, Ludwig Schmidt, Achal Dave.*
The Thirteenth International Conference on Learning Representations (ICLR 2025). April 2025.

[7] Language Models Scale Reliably with Over-training and on Downstream Tasks
Samir Yitzhak Gadre, Georgios Smyrnis, Vaishaal Shankar, Suchin Gururangan, Mitchell Wortsman, Rulin Shao, Jean Mercat, Alex Fang, Jeffrey Li, ***Sedrick Keh****, Rui Xin, Marianna Nezhurina, Igor Vasiljevic, Jenia Jitsev, Luca Soldaini, Alexandros G. Dimakis, Gabriel Ilharco, Pang Wei Koh, Shuran Song, Thomas Kollar, Yair Carmon, Achal Dave, Reinhard Heckel, Niklas Muennighoff, Ludwig Schmidt.*
The Thirteenth International Conference on Learning Representations (ICLR 2025). April 2025.

[8] Improving Test-Time Search for LLMs with Backtracking Against In-Context Value Verifiers
Anikait Singh, Kushal Arora, ***Sedrick Keh****, Jean Mercat, Tatsunori Hashimoto, Chelsea Finn, Aviral Kumar.*
Workshop on Reasoning and Planning for Large Language Models (ICLR 2025). April 2025.

[9] DataComp-LM: In search of the next generation of training sets for language models
Jeffrey Li, Alex Fang, Georgios Smyrnis, Maor Ivgi, Matt Jordan, Samir Gadre, Hritik Bansal, Etash Guha, ***Sedrick Keh****, Kushal Arora, Saurabh Garg, Rui Xin, Niklas Muennighoff, Reinhard Heckel, Jean Mercat, Mayee Chen, Suchin Gururangan, Mitchell Wortsman, Alon Albalak, Yonatan Bitton, Marianna Nezhurina, Amro Abbas, Cheng-Yu Hsieh, Dhruba Ghosh, Josh Gardner, Maciej Kilian, Hanlin Zhang, Rulin Shao, Sarah Pratt, Sunny Sanyal, Gabriel Ilharco, Giannis Daras, Kalyani Marathe, Aaron Gokaslan, Jieyu Zhang, Khyathi Chandu, Thao Nguyen, Igor Vasiljevic, Sham Kakade, Shuran Song, Sujay Sanghavi, Fartash Faghri, Sewoong Oh, Luke Zettlemoyer, Kyle Lo, Alaaeldin El-Nouby, Hadi Pouransari, Alexander Toshev, Stephanie Wang, Dirk Groeneveld, Luca Soldaini, Pang Wei Koh, Jenia Jitsev, Thomas Kollar, Alexandros G. Dimakis, Yair Carmon, Achal Dave, Ludwig Schmidt, Vaishaal Shankar.*
The Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024), Datasets and Benchmarks Track. December 2024.

[10] A Critical Evaluation of AI Feedback for Aligning Large Language Models
Archit Sharma, ***Sedrick Keh****, Eric Mitchell, Chelsea Finn, Kushal Arora, Thomas Kollar.*
The Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024). December 2024.

[11] SEACrowd: A Multilingual Multimodal Data Hub and Benchmark Suite for Southeast Asian Languages
Holy Lovenia, Rahmad Mahendra, Salsabil Maulana Akbar, Lester James V. Miranda, Jennifer Santoso, Elyanah Aco, Akhdan Fadhilah, Jonibek Mansurov, Joseph Marvin Imperial, Onno P. Kampman, Joel Ruben Antony Moniz, Muhammad Ravi Shulthan Habibi, Frederikus Hudi, Railey Montalan, Ryan Ignatius, Joanito Agili Lopo, William Nixon, Börje F. Karlsson, James Jaya, Ryandito Diandaru, Yuze Gao, Patrick Amadeus, Bin Wang, Jan Christian Blaise Cruz, Chenxi Whitehouse, Ivan Halim Parmonangan, Maria Khelli, Wenyu Zhang, Lucky Susanto, Reynard Adha Ryanda, Sonny Lazuardi Hermawan, Dan John Velasco, Muhammad Dehan Al Kautsar, Willy Fitra Hendria, Yasmin Moslem, Noah Flynn, Muhammad Farid Adilazuarda, Haochen Li, Johanes Lee, R. Damanhuri, Shuo Sun, Muhammad Reza Qorib, Amirbek Djanibekov, Wei Qi Leong, Quyet V. Do, Niklas Muennighoff, Tanrada Pansuwan, Ilham Firdausi Putra,

*Yan Xu, Ngee Chia Tai, Ayu Purwarianti, Sebastian Ruder, William Tjhi, Peerat Limkonchotiwat, Alham Fikri Aji, **Sedrick Keh**, Genta Indra Winata, Ruochen Zhang, Fajri Koto, Zheng-Xin Yong, Samuel Cahyawijaya.*
The 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP 2024). November 2024.

[12] Linearizing Large Language Models
Jean Mercat, Igor Vasiljevic*, **Sedrick Keh***, Kushal Arora, Achal Dave, Adrien Gaidon, Thomas Kollar.*
Conference on Language Modeling (COLM 2024). October 2024.

[13] Asking More Informative Questions for Grounded Retrieval
***Sedrick Keh**, Justin T. Chiu, Daniel Fried.*
Findings of the Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL Findings 2024). June 2024.

[14] Where It Really Matters: Few-Shot Environmental Conservation Media Monitoring for Low-Resource Languages
*Sameer Jain, **Sedrick Scott Keh**, Shova Chettri, Karun Dewan, Pablo Izquierdo, Johanna Prussman, Pooja Shrestha, César Suárez, Zheyuan Ryan Shi, Lei Li, Fei Fang.*
The 38th Annual AAAI Conference on Artificial Intelligence (AAAI 2024). February 2024.

[15] Doolittle: Benchmarks and Corpora for Academic Writing Formalization
*Shizhe Diao, Yongyu Lei, Liangming Pan, Tianqing Fang, Wangchunshu Zhou, **Sedrick Keh**, Min-Yen Kan, Tong Zhang.*
Conference on Empirical Methods in Natural Language Processing 2023 (EMNLP 2023). December 2023.

[16] PANCETTA: Phoneme Aware Neural Completion to Elicit Tongue Twisters Automatically
***Sedrick Scott Keh**, Steven Y. Feng, Varun Gangal, Malihe Alikhani and Eduard Hovy.*
Conference of the European Chapter of the Association for Computational Linguistics (EACL 2023). May 2023.

[17] Hashtag-Guided Low-Resource Tweet Classification
Shizhe Diao, **Sedrick Scott Keh***, Liangming Pan, Zhiliang Tian, Yan Song and Tong Zhang.*
International World Wide Web Conference 2023 (WWW 2023). May 2023.

[18] NewsPanda: Media Monitoring for Timely Conservation Action
Sedrick Scott Keh, Zheyuan Ryan Shi*, David J. Patterson, Nirmal Bhagabati, Karun Dewan, Areendran Gopala, Pablo Izquierdo, Debojyoti Mallick, Ambika Sharma, Pooja Shrestha, and Fei Fang.*
Conference on Innovative Applications of Artificial Intelligence (IAAI 2023). February 2023.

[19] EUREKA: EUphemism Recognition Enhanced through Knn-based methods and Augmentation
Sedrick Scott Keh, Rohit K. Bharadwaj*, Emmy Liu, Simone Tedeschi, Varun Gangal and Roberto Navigli.*
The Third Workshop on Figurative Language Processing (EMNLP 2022). December 2022.

[20] Exploring Euphemism Detection in Few-Shot and Zero-Shot Settings
***Sedrick Scott Keh**.*
The Third Workshop on Figurative Language Processing (EMNLP 2022). December 2022.

[21] PINEAPPLE: Personifying INanimate Entities by Acquiring Parallel Personification Data for Learning Enhanced Generation
***Sedrick Scott Keh**, Kevin Lu, Varun Gangal, Steven Y. Feng, Harsh Jhamtani, Malihe Alikhani and Eduard Hovy.*
International Conference on Computational Linguistics (COLING 2022). October 2022.

Preprints and Papers in Submission

[22] VLA Foundry: A Unified Framework for Training Vision-Language-Action Models
*Jean Mercat, **Sedrick Keh**, Kushal Arora, Isabella Huang, Paarth Shah, Haruki Nishimura, Shun Iwase, Katherine Liu.*
arXiv:2604.19728. April 2026.

[23] Semi-Supervised Noisy Student Pre-training on EfficientNet Architectures for Plant Pathology Classification
***Sedrick Scott Keh**.*
arXiv:2012.00332. December 2020.

[24] Myers-Briggs Personality Classification and Personality-Specific Language Generation Using Pre-trained Language Models
***Sedrick Scott Keh**, I-Tsun Cheng.*
arXiv:1907.06333. July 2019.

[25] MagmaDNN: Towards High-Performance Deep Learning Using Magma
*Daniel Nichols, **Sedrick Scott Keh**, Kam Fai Chan.*
Technical Report. July 2019.

Education

- **Carnegie Mellon University** — Pittsburgh, PA
  *MS in Machine Learning, GPA: **4.08/4.33*** — *Sept. 2021 – Dec. 2022*
- **Hong Kong University of Science and Technology** — Hong Kong
  *BSc in Data Science; BSc in Mathematics – First Class Honors, GPA: **4.09/4.30*** — *Sept. 2017 – Jun. 2021*

Awards and Scholarships

- **Scholarship** – HKUST Full Undergraduate Scholarship ($560,000 HKD) — 2017-2021
- **Scholarship** – S.S. Chern Class Mathematics Achievement Scholarship ($30,000 HKD) — 2018-2021
- **Finalist (Top 20; training pool for IMO)** – Philippine Mathematical Olympiad — 2015, 2016, 2017
- **Finalist (Top 30; training pool for IOI)** – National Olympiad in Informatics — 2017

Academic Service and Teaching

- **Reviewer:** NeurIPS (2026), ICLR (2025, 2026), COLM (2024, 2025, 2026), ACL (2023, 2024), EMNLP (2022, 2023, 2024), NAACL (2024), ARR (2023, 2024, 2025), CoNLL (2024, 2025, 2026), FigLang Workshop (2022, 2024), DL4C Workshop (ICLR 2025, NeurIPS 2025)
- **Conference Volunteer:** EMNLP 2022 (Abu Dhabi, UAE)
- **Teaching Assistant:** 10-617 Intermediate Deep Learning (Graduate-level) — Fall 2022
- **Teaching Assistant:** 10-315 Advanced Introduction to Machine Learning — Spring 2022
  - Nominated for the 2022 Alan J. Perlis Graduate Student Teaching Award
- **Mentor:** CMU Undergraduate AI Mentorship Program